RTB GIS ↔ TDMP Integration Requirements

What we need from the TDMP team · Prepared by the RTB GIS development team · 11 November 2026

RTB GIS (the TVET 3D GIS platform) will use **TDMP as the only source of truth for users and school information**. Users will sign in with their TDMP account, and RTB GIS will no longer hold its own user accounts or passwords. RTB GIS keeps its own sessions and audit log. **GPS coordinates** are the one school field RTB GIS keeps itself, because they are calibrated to our drone and 3D survey data.

How it will work

1. **Sign-in:** RTB GIS sends the user to `/oauth/authorize` (authorization code + PKCE). Our server exchanges the code at `/oauth/token`, then reads `/auth/me/access-scope` to get the user's role and scope.
2. **School data:** our server reads `/schools`, `/school-infrastructure` and `/teachers` with an API key, every night and on demand.
3. **From TDMP:** an "Open in RTB GIS" link on a TDMP school page signs the user in and opens that school directly.

1. Login (OAuth / OpenID Connect)

- [ ] Register an OAuth client for us (`POST /oauth/clients`):
  - name: `RTB GIS`
  - grants: `authorization_code`, `refresh_token`
  - redirect URIs: `https://rtb.aerovyntech.com/api/v1/auth/tdmp/callback` and `http://localhost:5173/api/v1/auth/tdmp/callback` (development)
- [ ] Turn on refresh tokens for this client.
- [ ] Skip the consent screen for this client, so users opening RTB GIS from TDMP are signed in straight away.
- [ ] Confirm that the access token from this login works on `/auth/me`, `/auth/me/access-scope` and `/auth/me/permissions`.
- [ ] Confirm that `email_verified` is reliable. We use the email to link existing RTB GIS accounts.
- [ ] Tell us the access-token and refresh-token lifetimes.
- [ ] Is there a sign-out endpoint (`end_session_endpoint`)? If not, how can we learn quickly that a user was deactivated?

2. Users and roles

- [ ] Send the list of TDMP roles and what each one may see, so we can map them to RTB GIS roles. Tell us which roles should have access to RTB GIS.
- [ ] Do all RTB headquarters staff and district officers have TDMP accounts, not only teachers?
- [ ] Send the full shape of `accessScope` in `/auth/me/access-scope`. For province- and district-level users we need to know *which* province or district.

3. API key (server-to-server data)

- [ ] A real API key from `POST /integrations/partners` for "RTB GIS". The documented example `fortress_sk_123` returns 401.
- [ ] Confirm the key can read `/schools`, `/schools/{schoolCode}`, `/school-infrastructure/{schoolCode}`, `/teachers` and `/locations/*`.
- [ ] Confirm which header format to use: `Authorization: ApiKey …` (from your docs) or `X-API-Key` (from the OpenAPI spec).
- [ ] Tell us the rate limits and the recommended sync frequency.

- [ ] Is there an "updated since" filter on `/schools`, so we only fetch changes?

4. School data

- [ ] Is `schoolCode` permanent? Which identifier should we store as the link: `schoolCode` or `id`?
- [ ] Which schools should RTB GIS show: TVET only (`categoryGroup=TVET`) or all institution types?
- [ ] We need these fields, which the TDMP school record doesn't include today. Can you provide them, or tell us where they live?
  - school phone, email, website, address
  - head teacher (or confirm we should use `isHeadTeacher` from `/teachers`)
  - students by gender; teachers, admin staff and support staff by gender
  - each trade/programme with its number of students and capacity
  - land area (used and unused); access roads (number, condition)
- [ ] Does `/teachers` include admin and support staff, or teachers only (`staffCategory`)?

5. Linking from TDMP to RTB GIS

- [ ] Add an "Open in RTB GIS" link on TDMP school pages:
  `https://rtb.aerovyntech.com/sso/tdmp?school={schoolCode}`

6. Testing

- [ ] A staging/test TDMP environment, or test data we may use on production.
- [ ] Test accounts for each kind of user: national admin, province-level, district-level, school-level, and one with "deny" scope.
- [ ] A technical contact person on the TDMP side.

> **Please share secrets securely.** Send the OAuth client secret and the API key through a secure channel, not by chat or plain email. They will be stored only in the RTB GIS server configuration.